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Temporal Physical Unclonable Function (T-PUF):

A Time-Irreversible Hardware Security Architecture Beyond Mathematical Hardness Assumptions

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Abstract

We present the **Temporal Physical Unclonable Function (T-PUF)**, a hardware security architecture whose security guarantee derives entirely from the thermodynamic arrow of time rather than from computational hardness assumptions. Conventional cryptographic systems (RSA, ECC, blockchain-based schemes) rely on hiding a static numerical secret; once that secret is copied, the entire security model collapses. T-PUF eliminates the secret-storage requirement by anchoring identity to an unreproducible physical moment: the instantaneous entropy state $E(t_0)$ of a hardware device at generation time t_0, which is consumed upon sampling and can never be reconstructed. The resulting commitment is safe to publish openly — possession of the commitment confers no advantage to an adversary who cannot reverse time. We formalise three physical axioms, derive a hybrid security model (thermodynamic + post-quantum ZK-STARKs), specify a three-tier multi-source entropy sampling scheme (SRAM start-up noise, CMOS dark-frame scatter, gyroscope micro-jitter), and describe a passive self-destruct honeypot defence mechanism. A 3D-stacked sovereign hardware integration guide completes the architecture.

Keywords: *Physical Unclonable Function, Thermodynamic Security, Time-Irreversible Cryptography, Zero-Knowledge Proof, Post-Quantum Security, Sovereign Hardware, 3D-Stacked Chip, SRAM PUF*

1. Introduction

The fundamental weakness of number-theoretic cryptography is that its secrets are *perfectly copyable*. A 256-bit private key is a string of bits; once observed, it can be reproduced with zero loss. Physical Unclonable Functions (PUFs) partially address this by deriving keys from manufacturing process variation, but existing PUF designs are vulnerable to machine-learning modelling attacks on challenge-response pairs.

Nature offers a deeper principle. Two leaves grown from the same branch at the same time share identical DNA yet differ in every microscopic vein — because the physical process of growth is governed by thermodynamic noise that is unique to each moment in time and space. T-PUF internalises this observation: **security is grounded not in mathematical difficulty but in the physical impossibility of reversing time.**

2. Formal Definition and Physical Axioms

2.1 Definitions

A conventional PUF maps challenges to responses deterministically:

```
Standard PUF:  f_HW : C  →  R
Security basis: manufacturing process variation (spatial uniqueness only)
```

T-PUF extends the domain to the space-time continuum:

```
T-PUF commitment at generation time t■:
```

```
  commit(t■) = Hash( f_HW(C) ■ E(t■) ■ t■ )

where:
  C       — public random challenge vector
  t■      — absolute generation timestamp (anchored, unforgeable)
  E(t■)   — instantaneous physical entropy snapshot at t■
  f_HW    — hardware fingerprint function (Layer 0, inaccessible to software)
```

2.2 Three Physical Axioms

Axiom I — Time Irreversibility: $\forall\ t_1 > t_0 : E(t_1) \neq E(t_0)$. The microscopic thermodynamic state of any physical device evolves monotonically; no process can restore a prior state.

Axiom II — Entropy Sampling Inseparability: Any attempt to observe or clone $E(t_0)$ necessarily injects energy that alters the state being observed. Sampling is consumption; lossless copying does not exist.

Axiom III — Timestamp Unforgeability: commit(t_0) must be anchored to the universe's time axis via physical multi-party witnessing or spatial-geometric constraints, preventing an adversary from substituting a later state for t_0.

3. Hybrid Security Model

T-PUF adopts a two-layer security model:

- **Identity uniqueness and unclonability** rest on Axioms I–III (thermodynamic law — unconditional).
- **Zero-knowledge public verifiability** relies on post-quantum ZK-STARKs (lattice-based — computationally secure, quantum-resistant).

This separation is critical: even if an adversary broke the ZK argument (forging a proof), they still cannot clone the physical device. The two layers address orthogonal attack surfaces.

Security proof sketch:

```
Breaking T-PUF identity
  ■   Reconstructing E(t■) from commit(t■)
  ■   Reversing thermodynamic entropy increase
  ■   Violating the Second Law of Thermodynamics
  ■   Physically impossible
```

Strongest adversary model: the adversary possesses the target device, a full memory image, complete algorithmic knowledge, and unbounded compute. T-PUF remains secure because the adversary cannot travel to t_0.

4. Multi-Source Entropy Sampling

Within the critical 100 ms cold-boot window, three entropy streams are fused:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 ■              T-PUF Entropy Fusion                  ■
 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 ■  PRIMARY          ■  SECONDARY       ■  AMBIENT       ■
 ■  SRAM start-up    ■  CMOS dark-frame ■  Gyroscope     ■
 ■  bitmap           ■  scatter noise   ■  micro-jitter  ■
 ■  (fab variation)  ■  (thermal+light) ■  (spatial)     ■
 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
          ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                           ▼
                     BLAKE3 Hash
                           ▼
                       commit(t■)
```

SRAM start-up noise is the primary source because it simultaneously satisfies all three axioms: the initial bit-map is determined by manufacturing variation (spatial uniqueness), the precise power-on voltage transient (temporal uniqueness), and electromigration-driven drift over device lifetime (time irreversibility). Multi-source fusion requires an adversary to predict all three streams simultaneously, producing exponential security amplification.

5. Delayed Zero-Knowledge Verification Protocol

The protocol follows Path B (hardware-as-function): the device fingerprint f_{HW} is the function; $E(t_0)$ serves only as the generation witness and is destroyed immediately after commit production.

Step 1 Commit (t■)	User publishes commit(t■) openly. Raw entropy E(t■) is physically zeroed immediately.
Step 2 Challenge (t■)	Verifier issues random challenge C at any future time t■.
Step 3 Response (t■)	Device produces R = Ψ(C) using its resident hardware fingerprint. No reconstruction of E(t■) is required or possible.
Step 4 ZK Proof (t■)	Local ZK-STARKs engine generates proof π: the hardware producing R is the same physical entity that produced commit(t■) at t■.
Step 5 Verify	Any party verifies (commit, C, R, π) without accessing device internals or E(t■).

6. Passive Self-Destruct Honeypot Defence

Rather than attempting to exclude attackers, T-PUF weaponises the act of intrusion itself as a trigger for ordered physical collapse:

```
Intrusion Event
      ■
      ■■■ Level 1 (soft)  Debugger / illegal register access
      ■                   → Freeze commit derivation chain (R permanently invalidated)
      ■
      ■■■ Level 2 (hard)  Capacitance spike / JTAG probe / delamination
      ■                   → Active SRAM discharge (hardware fingerprint overwritten)
      ■
      ■■■ Level 3 (nuke)  Repeated unauthorised verification attempts
                          → Full local identity self-destruct (physical open-circuit)
```

On Level 1/2 trigger, the system returns a thermally-seeded pseudo-response R_{pseudo} that appears plausible, luring the adversary deeper while logging their behaviour as legally admissible physical evidence.

Legal note: all countermeasures are purely passive and contained within the owner's device. No outbound access to any third-party system occurs.

7. Sovereign Hardware Integration (PLC-STD-001)

```
Layer 3  Application API      ZK-STARKs proof engine, open interface
  Layer 2  Commitment layer     BLAKE3 hardware-accelerated hash unit
  Layer 1  Entropy sampler      100 ms cold-boot: SRAM + CMOS + IMU
  Layer 0  Physical PUF core    Transistor mismatch silicon fingerprint
                                No software access / No JTAG
```

3D-stacked chip integration mapping:

```
chip3d_security_top  ■■■  Layer 3 + Layer 2  (ZKP & hash engines)
        ▲
   tsv_channel          ■■■  Dedicated entropy transport via TSV
        ▼
  sovereign_base        ■■■  Layer 1 + Layer 0  (SRAM cells + sensors)
        ▲
  m2_intercept          ■■■  Self-destruct discharge gates
```

8. Comparison with Existing Security Systems

Dimension	RSA / AES	Blockchain	SRAM PUF	T-PUF (this work)
Security root	Math hardness	Consensus+math	Spatial uniqueness	**Thermodynamic law**
Key storage	Required	Required	Zero (dynamic)	**Zero (entropy destroyed)**
Public commitment	Fatal if leaked	Fatal if leaked	Not possible	**Safe to publish openly**
Quantum resistance	Broken (Shor)	Broken	Inherent	**Absolute (physics)**
Physical relocation	Trivially copied	Trivially dumped	Partial risk	**Immune (t■ has passed)**
Defence posture	Passive	Passive	Static	**Active self-destruct honeypot**

9. Open Research Problems

P1 — ZK proof latency

Optimise ZK-STARKs proof generation on mobile SoCs to complete within the 100 ms sampling window at acceptable thermal cost.

P2 — Entropy source ageing compensation

SRAM start-up bitmaps drift slowly due to electromigration. Design a fuzzy extractor with sufficient error tolerance to maintain commit alignment over a 5-year device lifetime.

P3 — Decentralised timestamp anchoring

Investigate joint anchoring via a decentralised time chain and physical quantum events (e.g. radioactive decay pulses) to eliminate trust in centralised NTP/GPS sources.

P4 — Sub-micron probe countermeasures

Model the discharge rate of m2_intercept capacitors under focused-ion-beam probing to guarantee self-destruct completes before the first charge extraction.

References

[1] Pappu, R. et al. Physical One-Way Functions. *Science* 297, 2026–2030 (2002).

[2] Maes, R. Physically Unclonable Functions: Constructions, Properties and Applications. Springer (2013).

[3] Ben-Sasson, E. et al. Scalable, transparent, and post-quantum secure computational integrity. *IACR ePrint* 2018/046 (2018).

[4] National Institute of Standards and Technology. SHA-3 Standard: BLAKE3. NIST FIPS 202 (2015).

[5] Mao, G. SiliconForge Security SoC V3.0 — Open Source Release. Zenodo. DOI: 10.5281/zenodo.sfsoc-v3.2025 (2025).